ARTIFICIAL INTELLIGENCE

Assignment – 1

AI-Based Cyber Attack Detection Agent

Submitted By

Vismaya P K

Group: 10

S7 CSE A

Roll No:61

AI-Based Cyber Attack Detection Agent

1.Introduction

The AI-Based Cyber Attack Detection Agent is an intelligent system that monitors network activities and detects suspicious or malicious behavior in real time. It analyzes network traffic, login attempts, and system logs using artificial intelligence techniques to identify cyber threats and generate timely alerts.

The system helps improve network security by detecting attacks quickly and supporting automatic responses such as blocking suspicious IP addresses and notifying administrators. This reduces security risks, minimizes response time, and enhances the overall reliability of computer networks.

2. PEAS Representation

Component	Description
Performance Measure	High attack detection accuracy, low false alarms, fast response time, improved network security, and reliable threat detection.
Environment	Computer networks, servers, routers, firewalls, cloud systems, user devices, network traffic, and system logs.
Actuators	Block malicious IP addresses, generate security alerts, update firewall rules, isolate infected devices, and generate security reports.
Sensors	Network traffic monitor, firewall logs, system logs, login records, packet analyzer, IDS/IPS logs, and user activity logs.

3. Environmental Properties

Property	Classification	Reason
Observability	Partially Observable	The agent cannot observe every activity in the network completely.
Agents	Single Agent	Only one AI agent performs cyber attack detection.
Deterministic / Stochastic	Stochastic	Cyber attacks occur unpredictably and involve uncertainty.
Episodic / Sequential	Sequential	The current decision depends on previous network activities and events.
Static / Dynamic	Dynamic	The network environment changes continuously with user activities and traffic.
Discrete / Continuous	Continuous	Network traffic and system activities occur continuously over time.
Known / Unknown	Partially Known	Some attack patterns are known, while new threats may be unknown.

4. Rationality Measures

The AI-Based Cyber Attack Detection Agent continuously monitors network traffic, system logs, login attempts, and user activities to identify suspicious behavior. It compares the collected data with normal network patterns using artificial intelligence techniques and determines whether an activity is normal or malicious.

When abnormal behavior such as repeated failed login attempts, malware activity, or unauthorized access is detected, the agent immediately generates alerts, blocks suspicious IP addresses, updates firewall rules, and prepares security reports. By making timely and accurate decisions, the agent improves network security, reduces response time, and minimizes the impact of cyber attacks.

The agent continuously learns from newly detected threats and adapts its detection strategy to improve future performance. This intelligent decision-making process helps maintain a secure network environment, reduces manual monitoring efforts, and enhances the overall efficiency of cybersecurity management.

5. Innovation Point

The proposed AI-Based Cyber Attack Detection Agent includes several innovative features that improve the efficiency and accuracy of cybersecurity systems.

1. **Self-Learning Threat Detection:** The agent continuously learns from new cyber attack patterns and improves its detection accuracy over time using artificial intelligence.
2. **Automatic Incident Response:** The system automatically blocks suspicious IP addresses, updates firewall rules, and generates security alerts without requiring immediate human intervention.
3. **Real-Time Monitoring Dashboard:** The agent provides a live dashboard that displays network status, detected threats, attack severity, and security reports, enabling administrators to respond quickly to cyber incidents.

6. Diagram

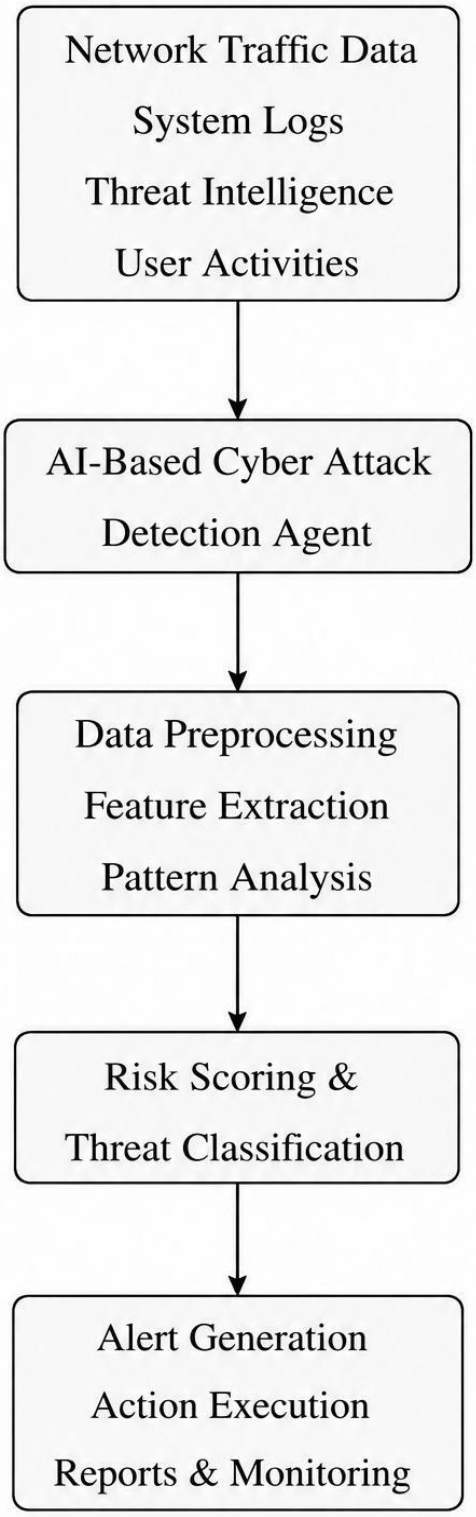


Fig 1 : Architecture of the AI -Based Cyber Attack Detection Agent

7. Conclusion

The AI-Based Cyber Attack Detection Agent provides an intelligent and efficient solution for detecting and responding to cyber threats in real time. By analyzing network traffic, system logs, and user activities, the agent can identify malicious behavior and take appropriate security actions.

The system improves network security by reducing response time, minimizing false alarms, and supporting automatic threat mitigation. Its ability to learn from new attack patterns makes it a reliable and effective solution for modern cybersecurity environments.

8. Simulation

The simulation of the **AI-Based Cyber Attack Detection Agent** demonstrates the working of the intelligent system in detecting potential cyber threats. The user enters network-related information such as the number of normal packets, failed login attempts, unknown IP connections, and malware detection status. The AI agent analyzes these inputs using predefined rules and calculates the threat level based on the detected suspicious activities.

If the calculated threat level exceeds the predefined threshold, the system identifies the activity as a cyber attack and immediately generates a security alert. It also recommends appropriate actions such as blocking suspicious IP addresses, updating firewall rules, and notifying the system administrator. If no suspicious activity is detected, the system reports that the network is safe and continues monitoring. This simulation demonstrates how the AI agent can provide quick and efficient threat detection while improving the overall security of the network.

9. Screenshot of Full Code

The following screenshots show the complete implementation of the AI – Based Cyber Attack Detection Agent.



Fig 2: Program Initialization and Input Module.

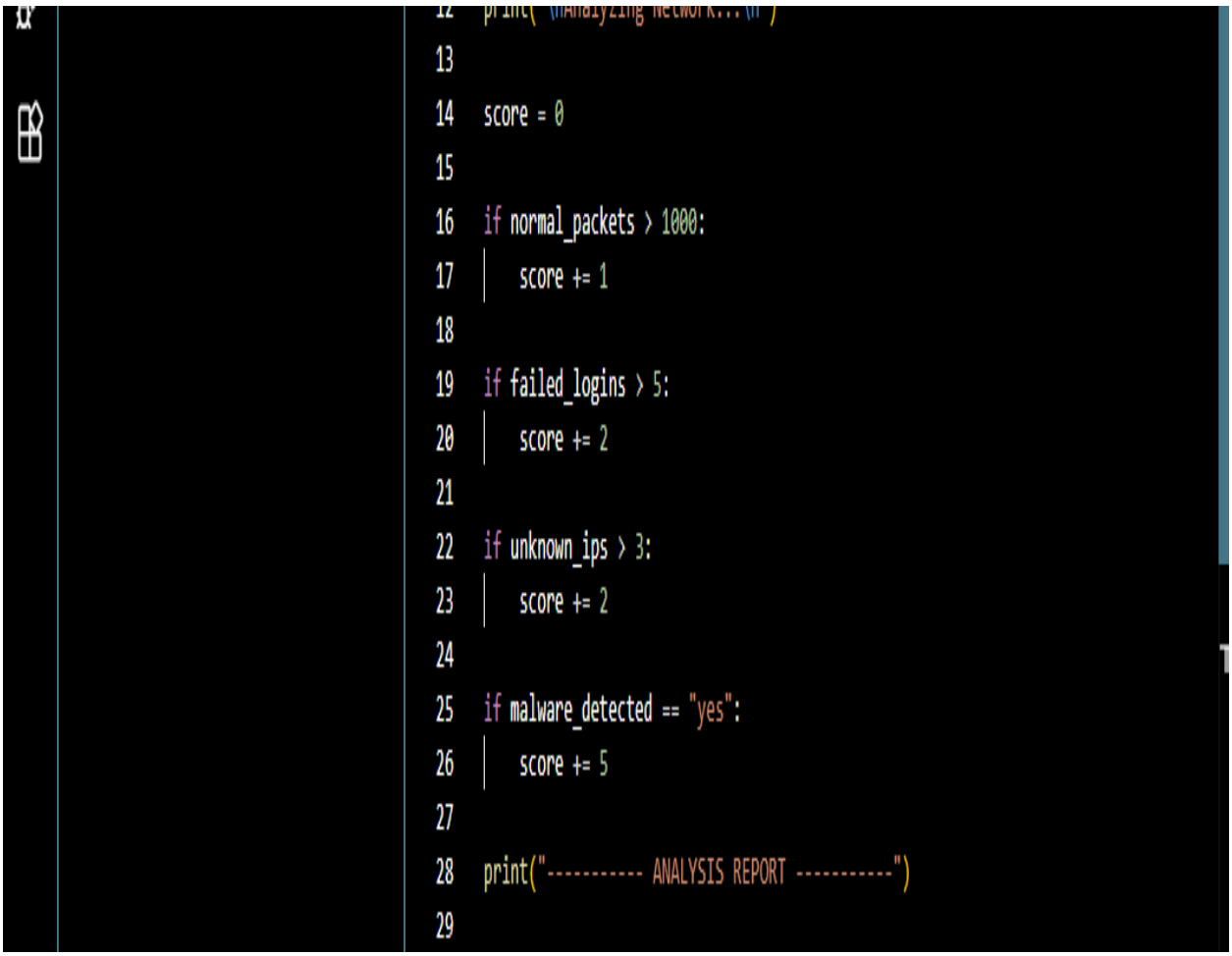


Fig 3 : AI -Based Attack Detection Algorithm.

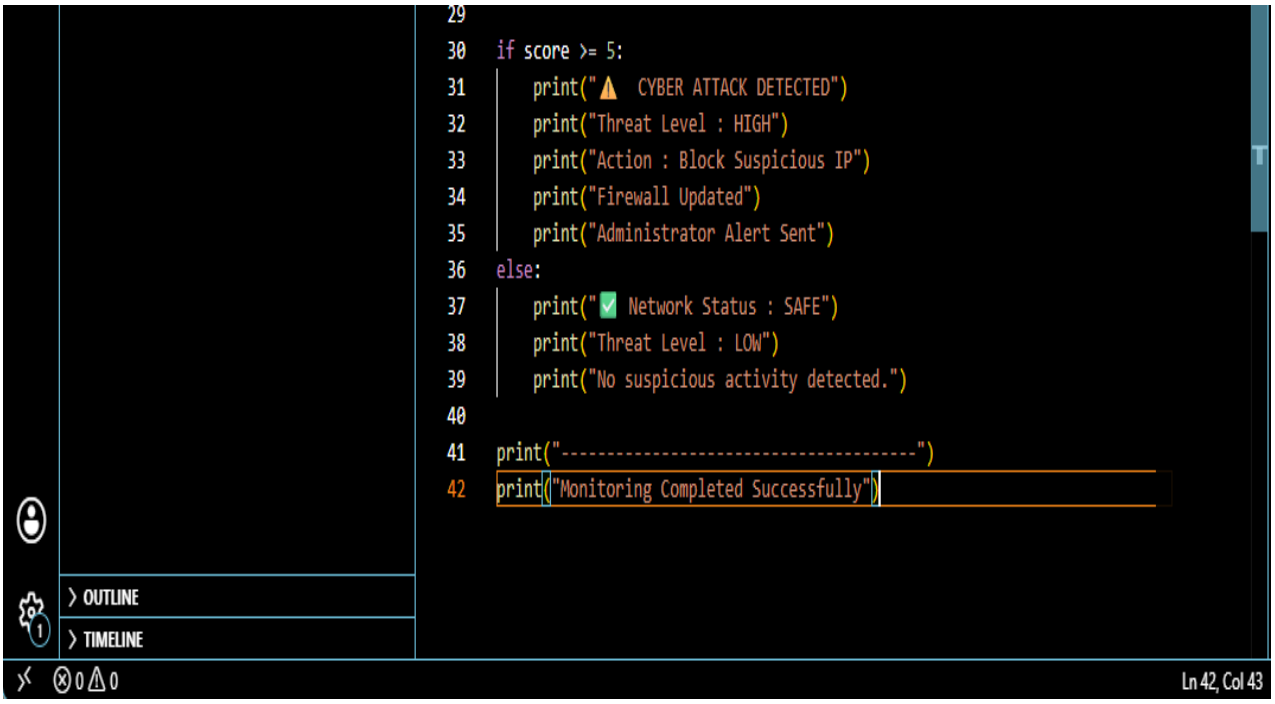


Fig 4: Output Generation Module.

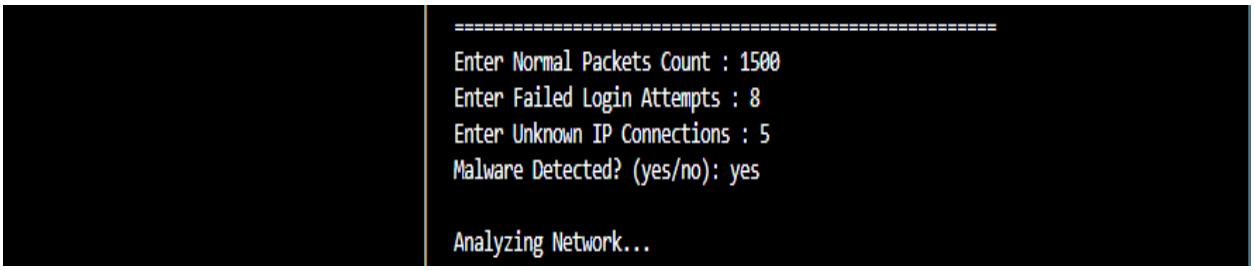


Fig 5 : Program Execution.

10. Output

The following show the execution of the program and the final output generated by the AI - Based Cyber Attack Detection Agent.



Fig 6: Attack Detected.

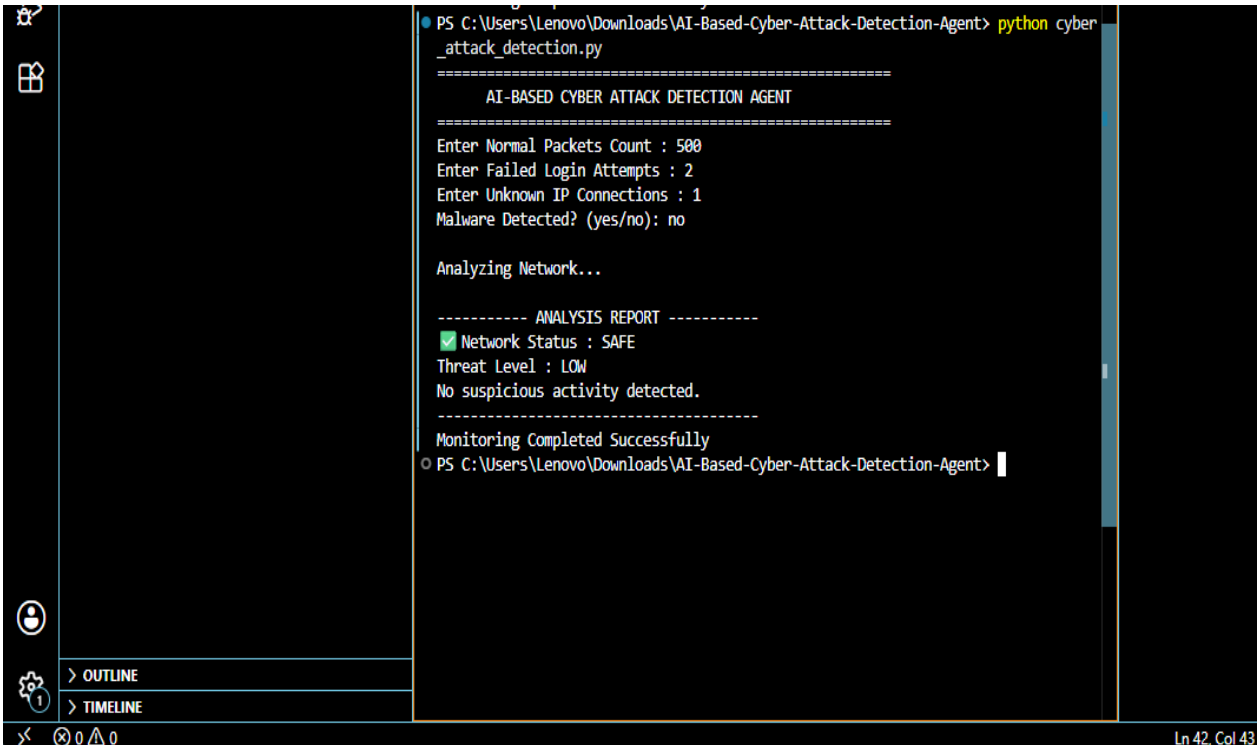


Fig 7 : Network Safe .